Glycolysis = breakdown of sugars; glycogen, glucose, fructose

Where in body?

Where in cell?

What are the inputs?

What are the outcomes?

Oxygen required?

Gibbs Free Energy Changes

Rxn#	Enzyme		ΔG°'(kJ/mol)	ΔG(kJ/mol)
1	Hexokinase	-16.7	-33.5	
2	Phosphogluco-isomerase		+1.7	-2.5
3	Phosphofructokinase		-14.2	-22.2
4	Aldolase		+23.9	-1.3
5	Triose phos. Isomerase		+7.6	+2.5
6	G-3-PDH	+12.6	-3.4	
7	Phosphoglycerate kinase		-37.6	+2.6
8	Phosphoglycerate mutas		+8.8	+1.6
9	Enolase		+3.4	-6.6
10	Pyruvate kinase	-62.8	-33.4	

Identify:

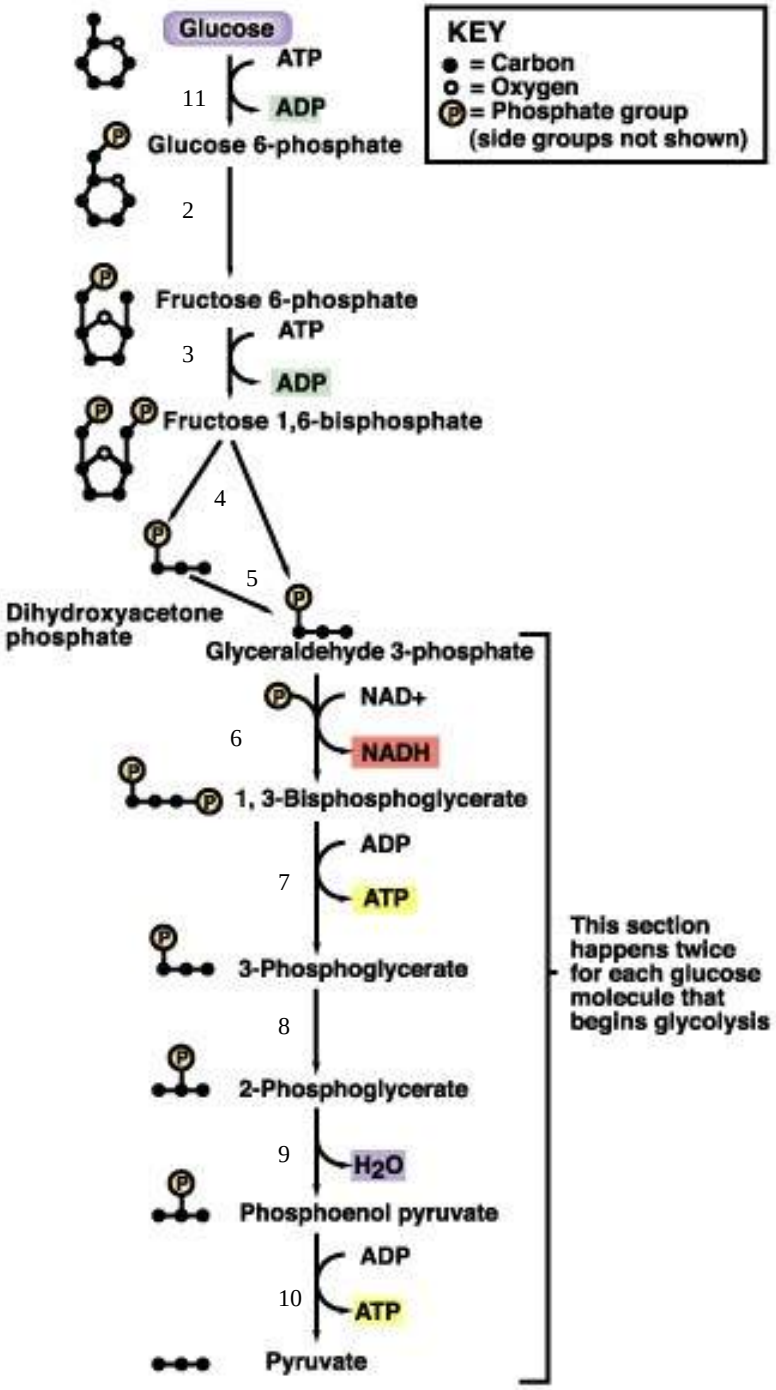
endergonic rxns

exergonic rxns

coupled reactions

oxidation/reduction rxns

transfer reactions



When do we use glycolysis?

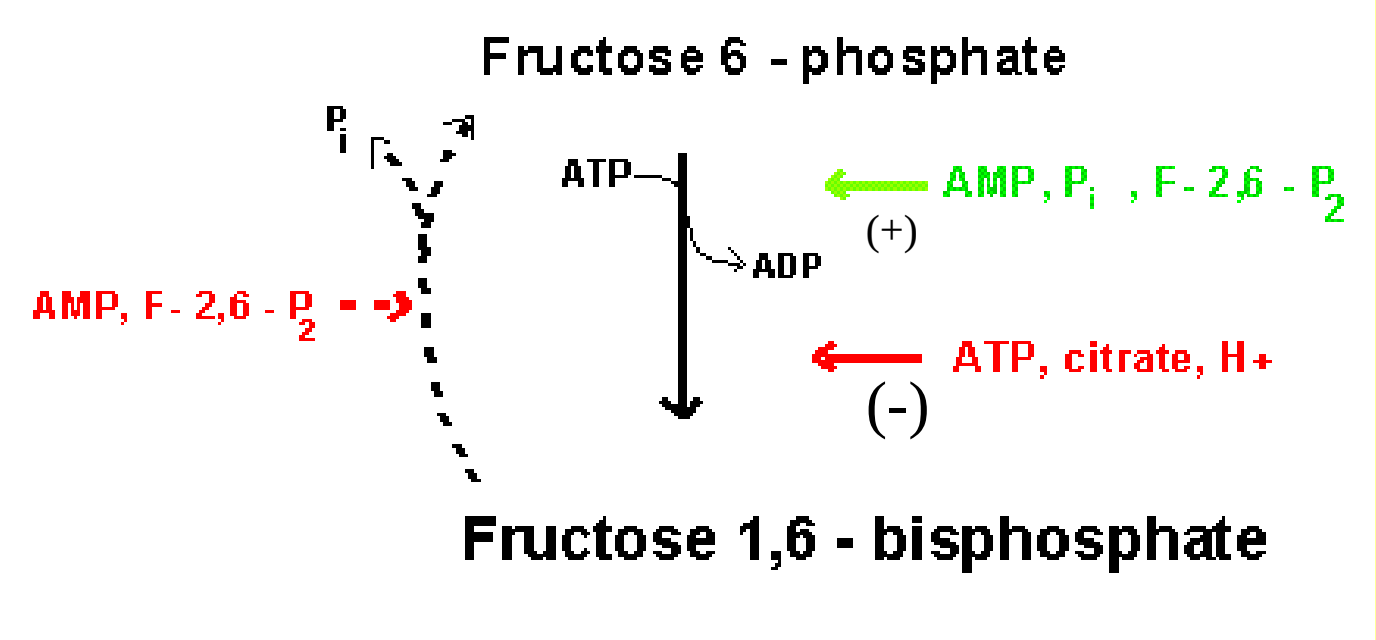
What are the advantages of using glycolysis for energy supply?

What are the disadvantages?

How is glycolysis regulated?

Hexokinase inhibited by glucose –6-phosphate; also there are several isoforms; lowest Km in liver

Phosphofructokinase (PFK)



Pyruvate kinase inhibited by ATP and acetylCoA; activated by fructose 1,6 bisphosphate

Where do the intermediates in glycolysis go?

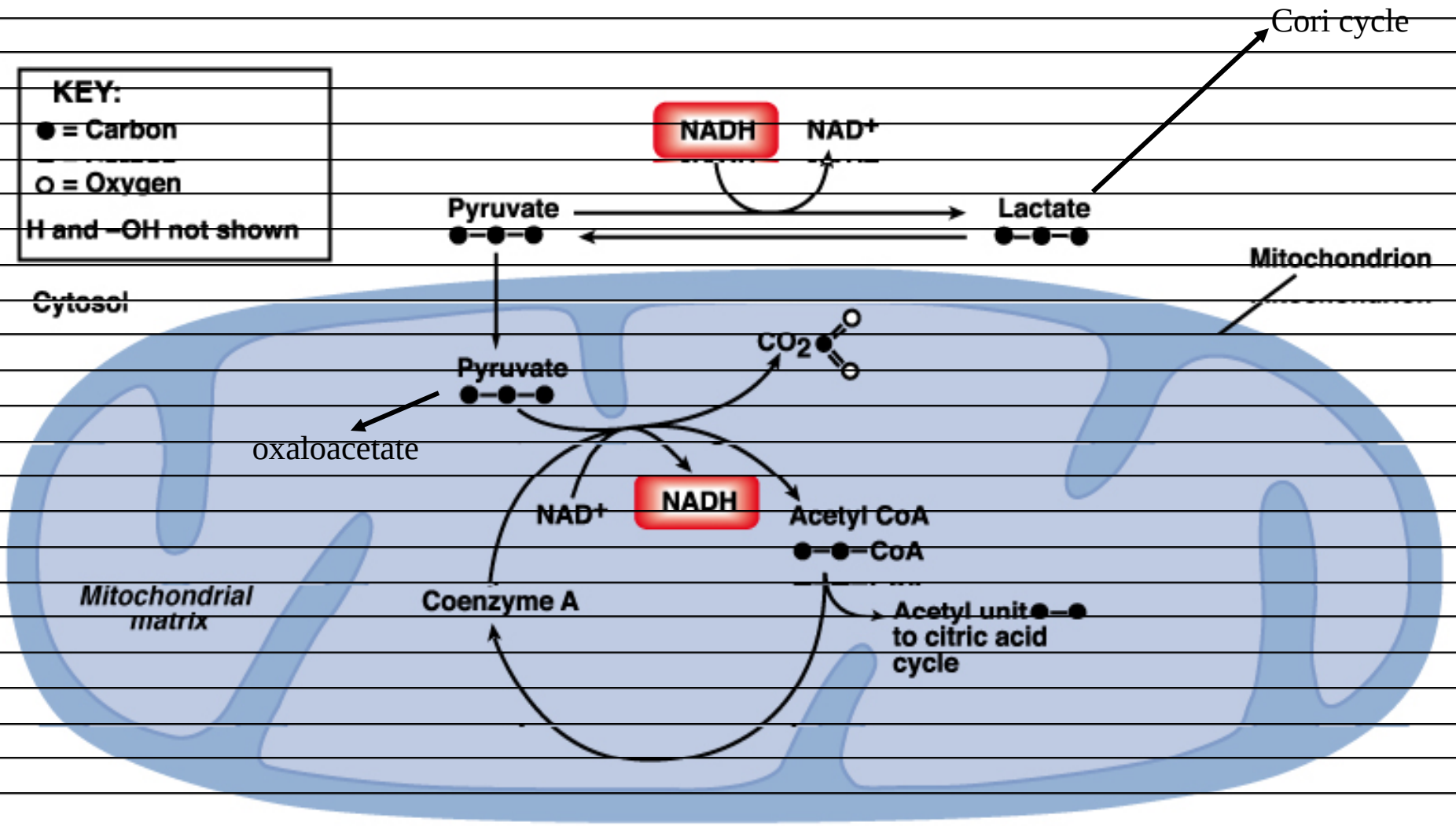
- G-6-P goes off to make the ribose for nucleotides
- F-6-P -amino sugars-glycolipids and glycoproteins
- G-3-P/DHAP-lipids
- 3PG-serine
- PEP-aromatic amino acids, pyrimidines, asp and asn
- Pyruvate-alanine

This pathway not only important in glucose metabolism--generates intermediates for other important building blocks

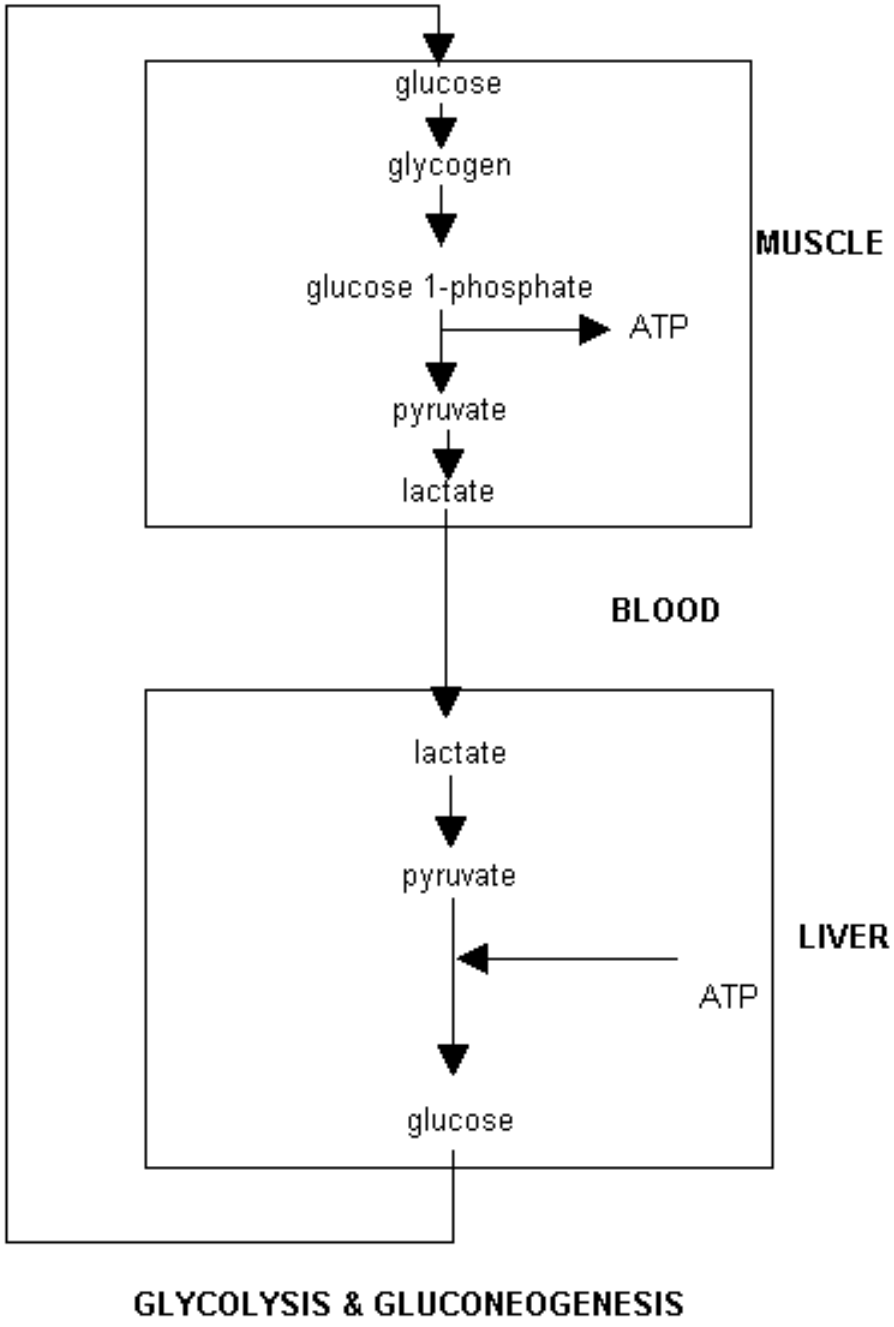
G-6-P = glucose 6 phosphate, F-6-P = fructose 6 phosphate, G-3-P = glyceraldehyde 3 phosphate, DHAP = dihydryoxacetonephosphate, 3PG = phosphoglyceraldehyde, Pyr = pyruvate

What are the possible fates of pyruvate?

- Ethanol (fermentation)
- Acetyl coA (mammals and others)
  - TCA/Krebs cycle
- Oxaloacetate - gluconeogenesis
- Lactate (mammals and others)
  - End product of anaerobic glycolysis
  - Gluconeogenesis in liver via the Cori cycle


Cori cycle
KEY:
● = Carbon
○ = Oxygen
H and –OH not shown
NADH
NAD+
Pyruvate
Lactate
Mitochondrion
Cytosol
Pyruvate
CO2
oxaloacetate
NAD+
NADH
Acetyl CoA
CoA
Mitochondrial matrix
Coenzyme A
Acetyl unit to citric acid cycle

Cori Cycle


glucose
glycogen
glucose 1-phosphate
ATP
pyruvate
lactate
MUSCLE
BLOOD
lactate
pyruvate
ATP
glucose
LIVER
GLYCOLYSIS & GLUCONEOGENESIS

Energy Balance Sheet for the Oxydation of Glucose via Glycolysis

Gains:

4 ATP

2 pyruvate

2 NADH + H+

Mitochondria for further oxidation via the TCA/Krebs cycle

Net Gain:

+ 2 ATP

Losses:

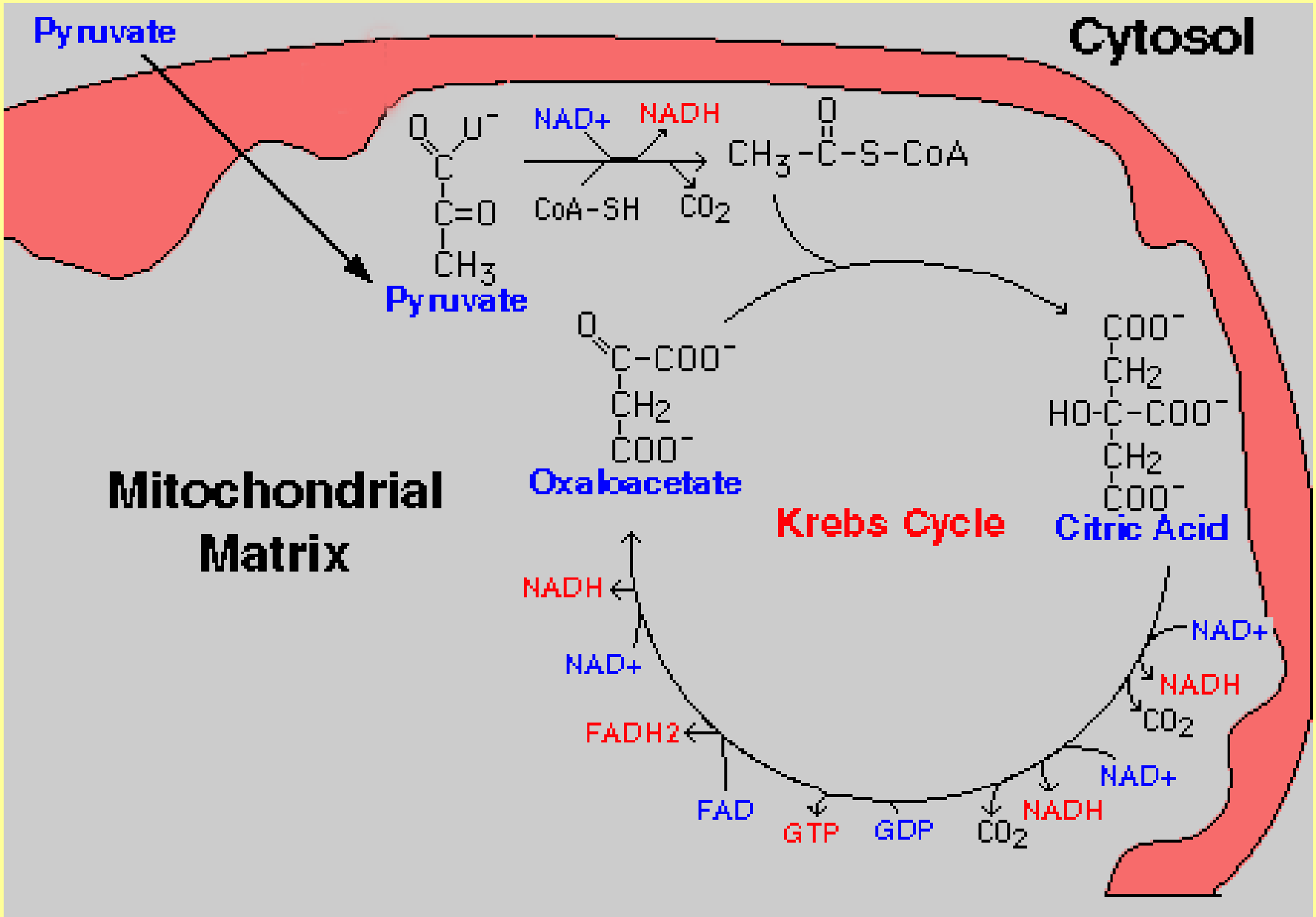
2ATP

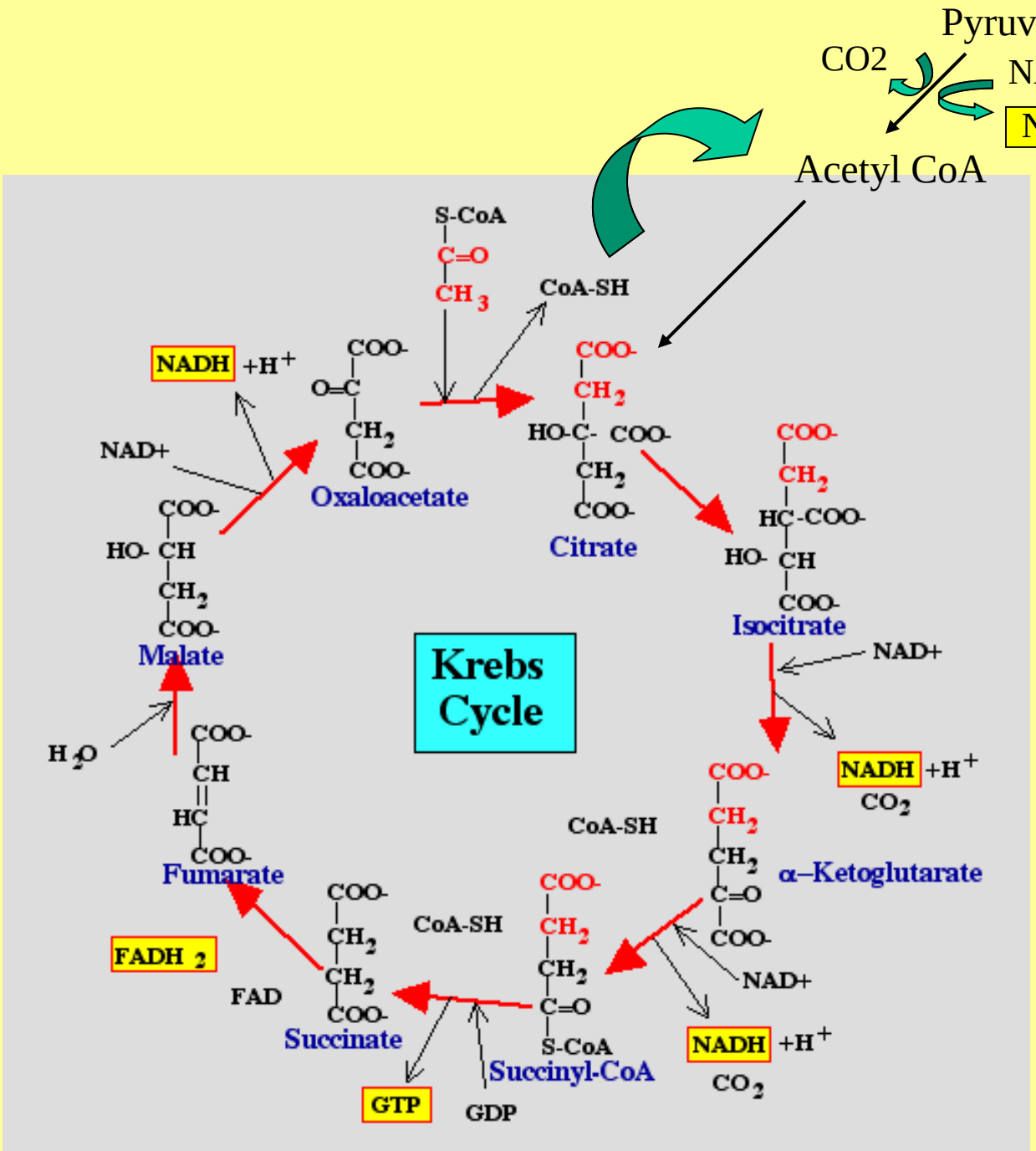
Glucose

Phosphate

NAD+ (recycled)

Oxidation of pyruvate via the TCA/Krebs/Citric Acid Cycle





- All compounds are tricarboxylic acids
- Carbons from glucose are shown in red
- Carbons from glucose are lost as CO2 (decarboxylation)
- Several NADH + H+ are generated via oxidation of intermediates
- One high energy phosphate compound (GTP)is produced

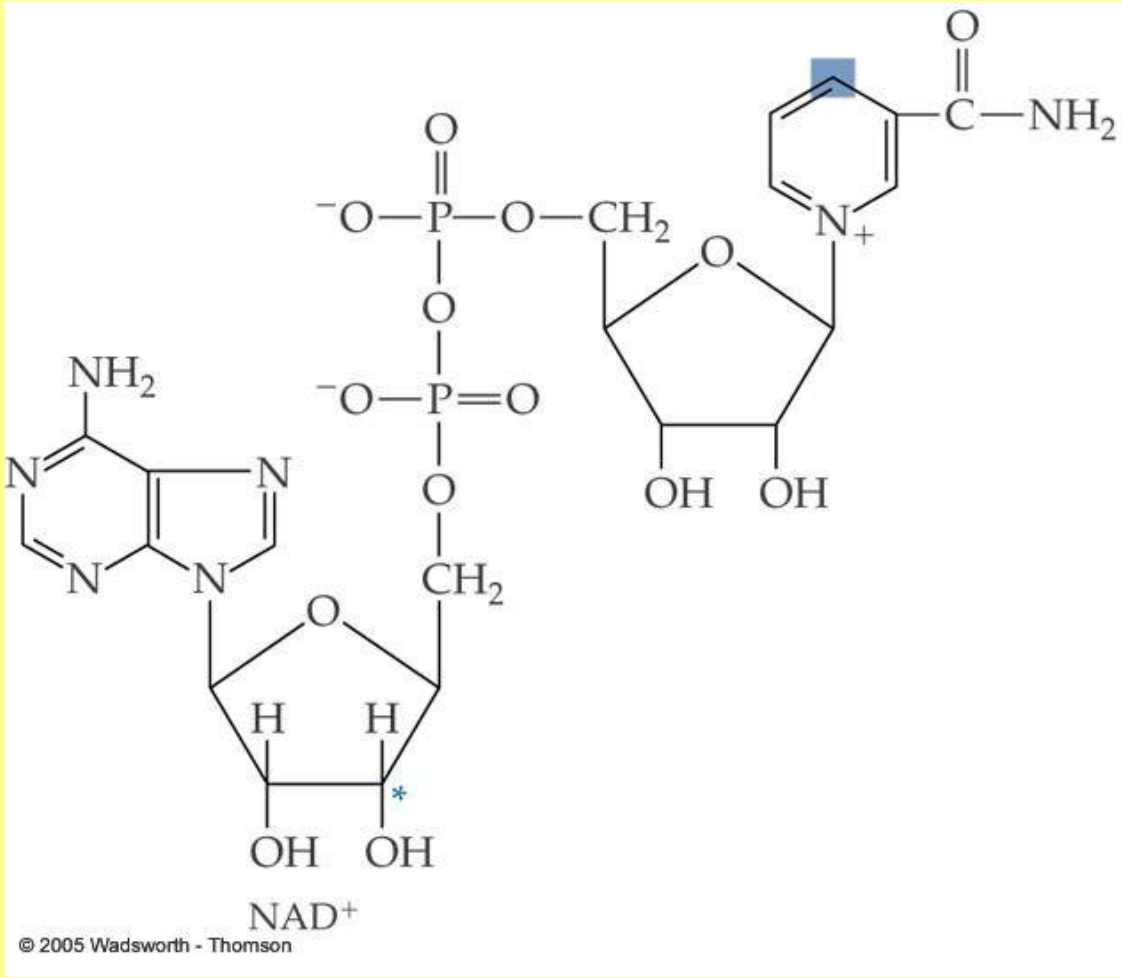
When do we oxidize pyruvate via the Krebs cycle?

What do we need to accomplish the oxidation of pyruvate?

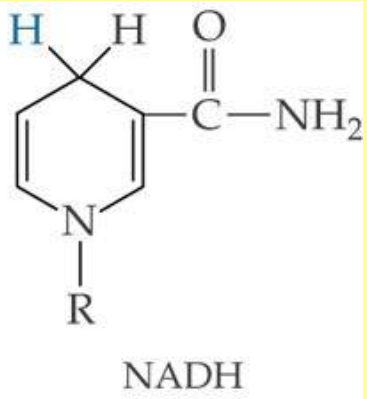
- NAD+ and FAD+; each can carry 2 e-
- oxygen; needs 2 e- to fill outer valence shell of electrons
- glucose

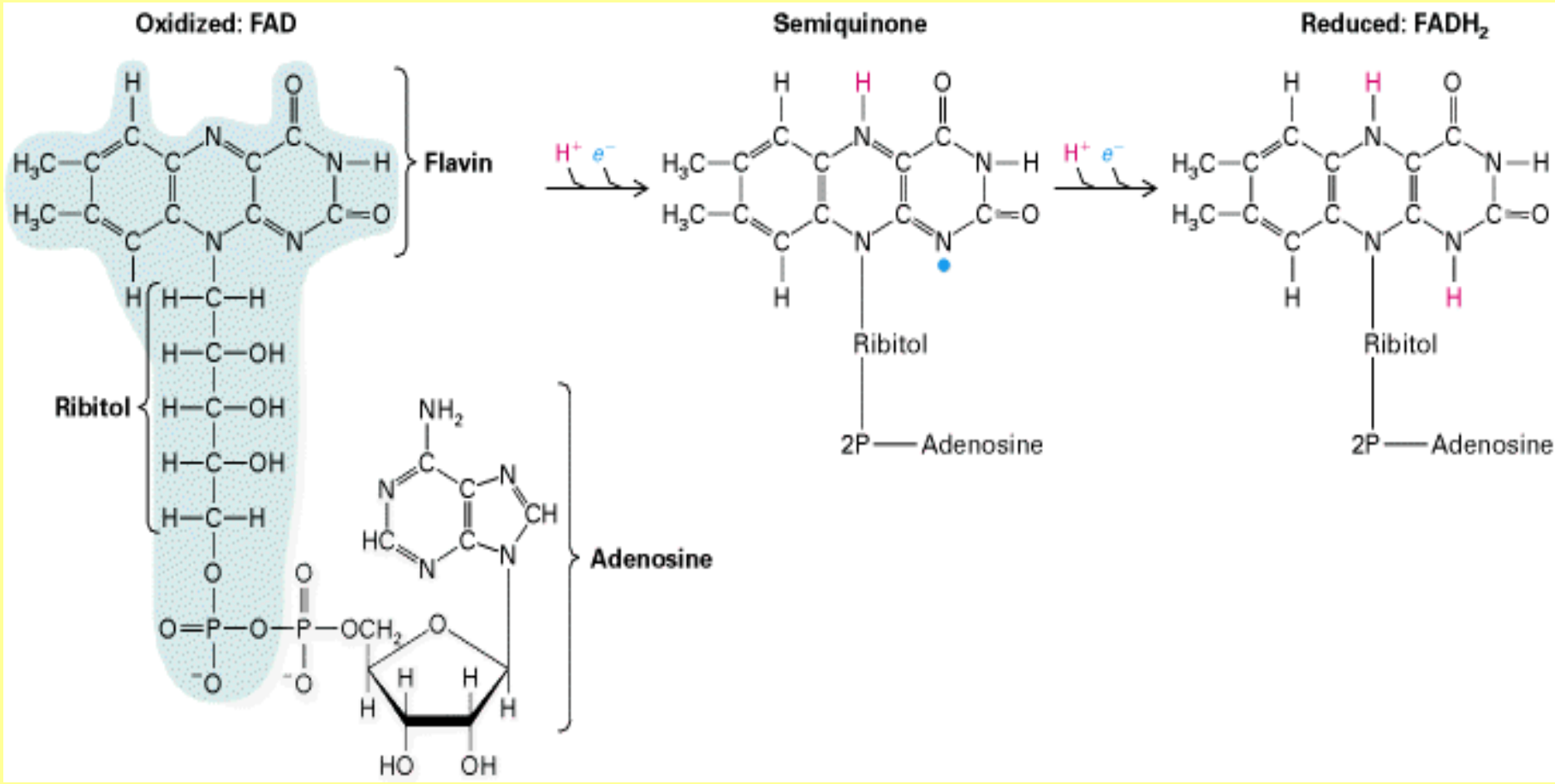
Where are the Krebs cycle enzymes and electron transport proteins located?

- Krebs cycle enzymes are located in the mitochondrial matrix
- Electron transport proteins in the inner mitochondrial membrane



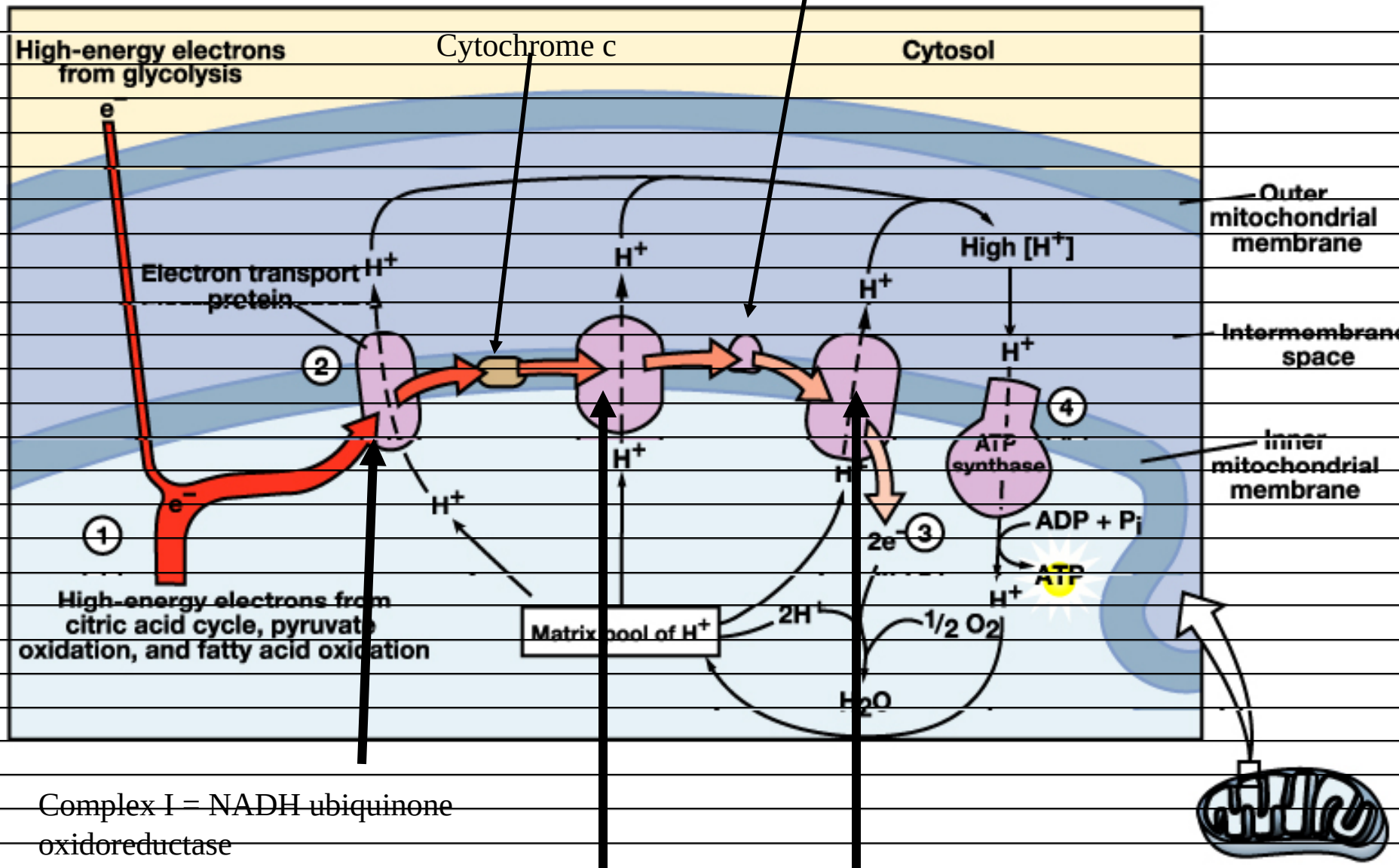
© 2005 Wadsworth - Thomson




Oxidized: FAD
Semiquinone
Reduced: FADH2
Flavin
Ribitol
Adenosine
H+ e−
H+ e−
Ribitol
2P — Adenosine
Ribitol
2P — Adenosine

Prosthetic groups = Fe, Flavin, Fe-S, Cu

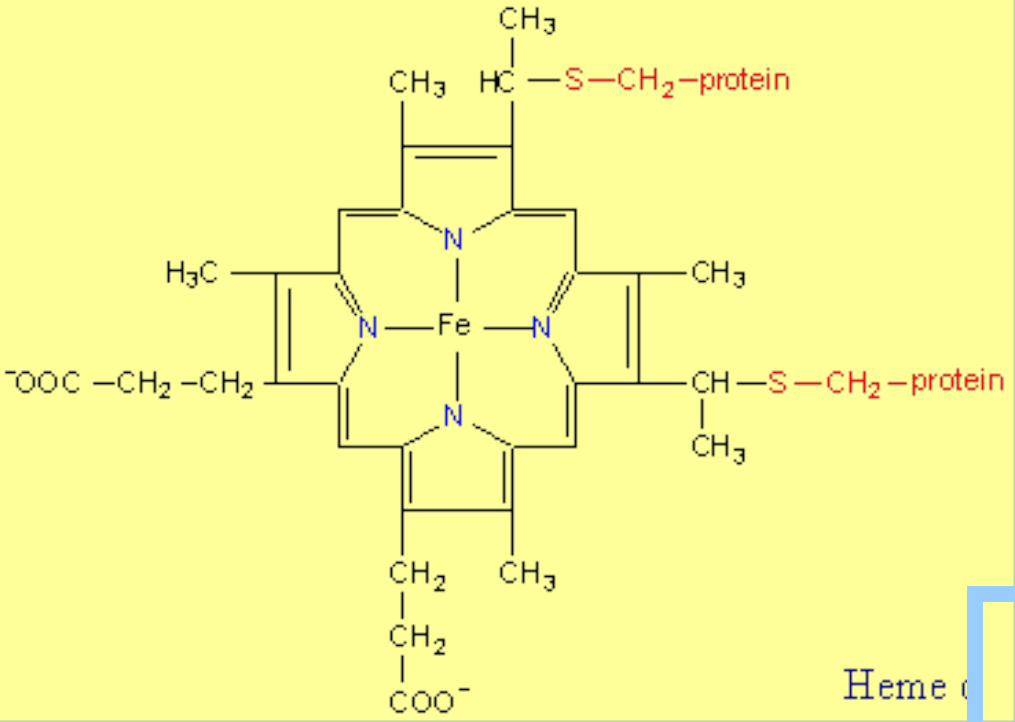
Coenzyme Q (ubiquinone)



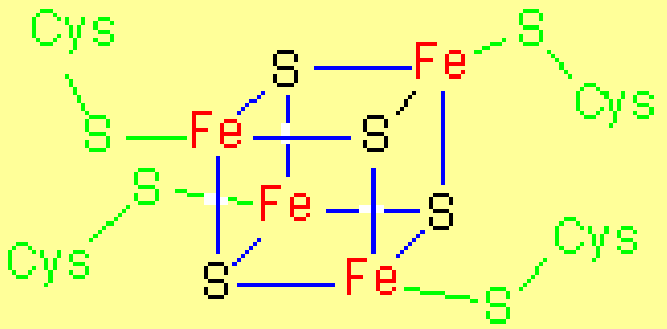
Complex I = NADH ubiquinone oxidoreductase

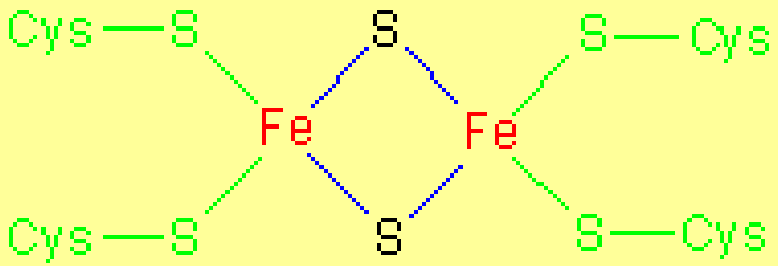
Complex II = succinate-ubiquinone oxidoreductase

Complex III = cytochrome c oxidoreductase



Heme

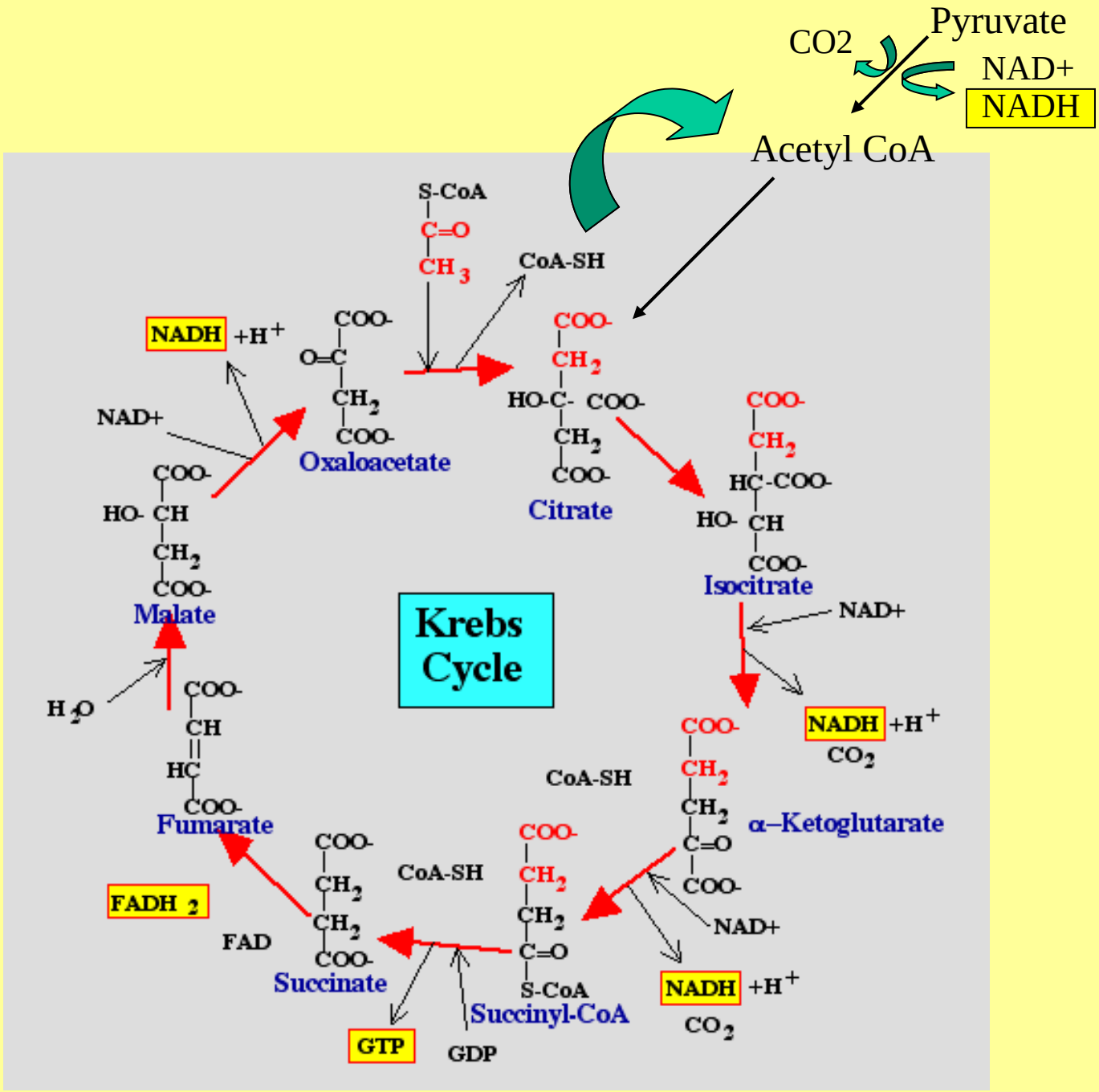


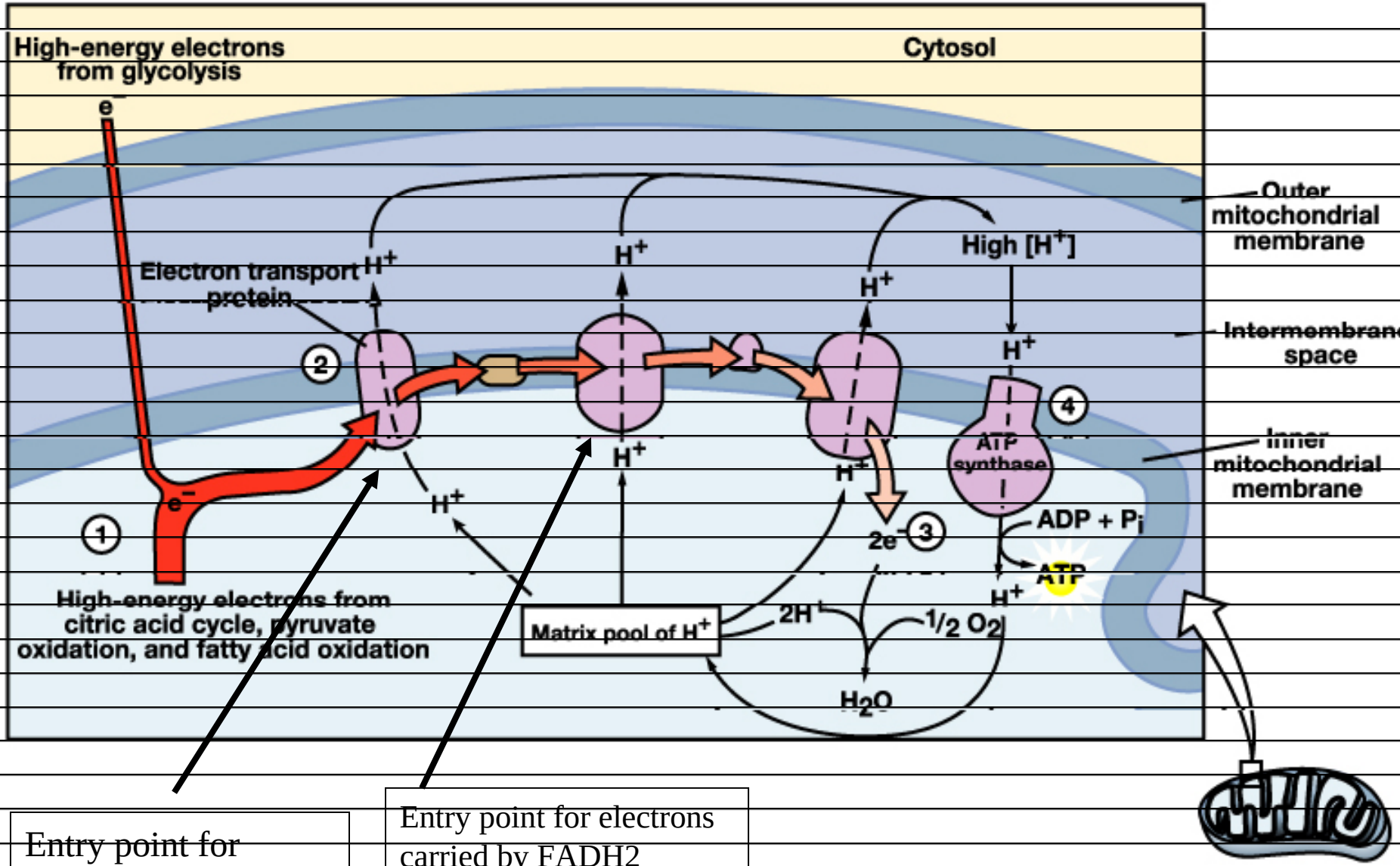


Iron-Sulfur Centers

Electron transport proteins

- each can accept or give up two electrons
- one protein in each complex also acts as a hydrogen pump
- electron entry point is determined by the energy state of the electrons


Pyruvate
CO2
NAD+
NADH
Acetyl CoA
S-CoA
C=O
CH 3
CoA-SH
NADH +H+
NAD+
Oxaloacetate
Citrate
Isocitrate
Malate
Krebs Cycle
NAD+
NADH +H+
CO2
H2O
Fumarate
CoA-SH
α–Ketoglutarate
FADH 2
FAD
CoA-SH
Succinate
Succinyl-CoA
NAD+
NADH +H+
CO2
GTP
GDP



Entry point for electrons carried by NADH+ H+

Entry point for electrons carried by FADH2

Net Energy Yield from the Oxidation of Pyruvate via the TCA cycle

From Glycolysis:

+2NADH +2ATP

+ +

From TCA:

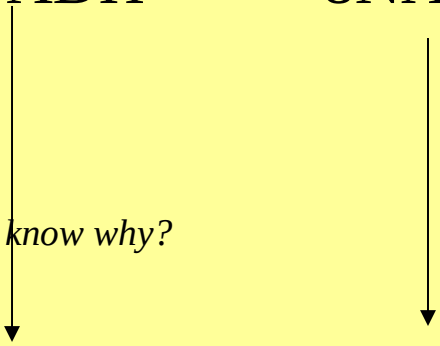
+2FADH +8NADH +2GTP

ETC:

3ATP/NADH

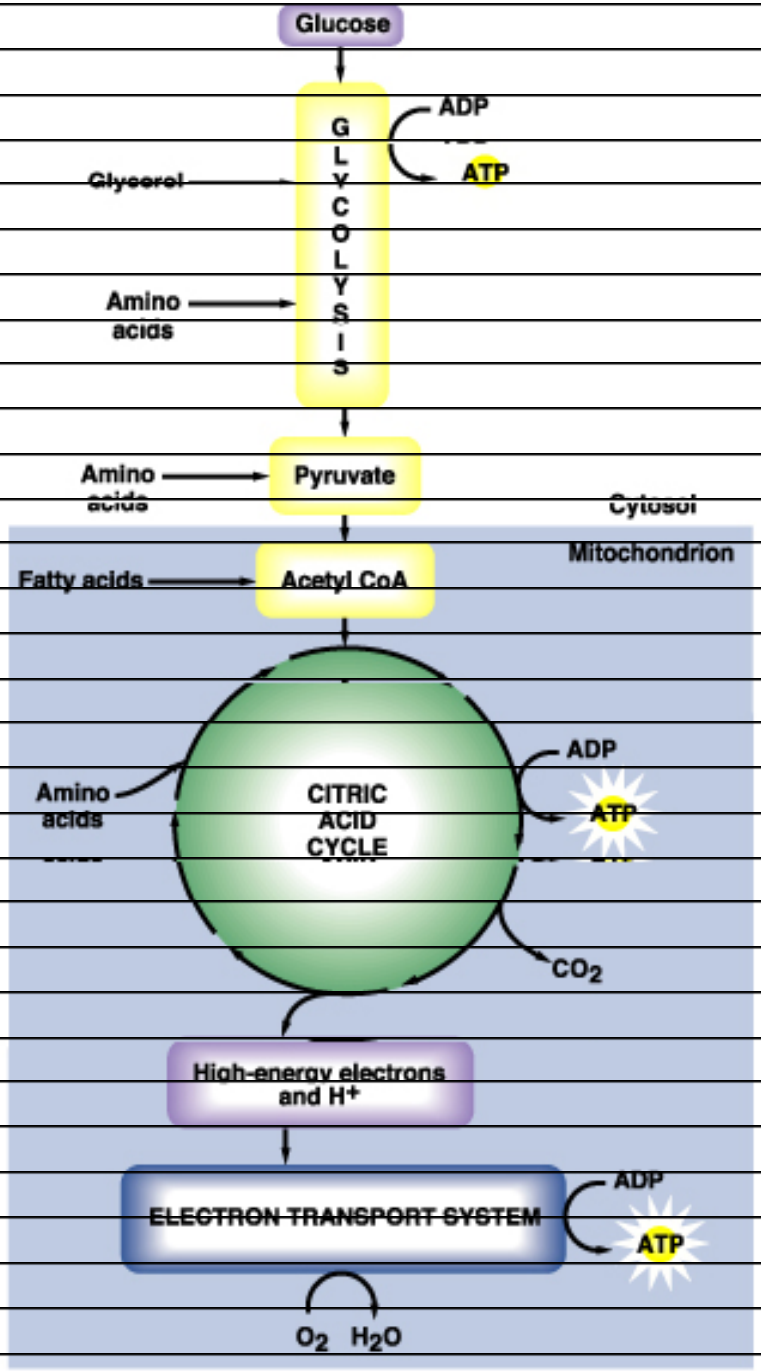
2ATP/FADH

Do you know why?



+4ATP +30ATP

+38ATP TOTAL


Glucose
ADP
ATP
Glycerol
GLYCOLYSIS
Amino acids
Amino acids
Pyruvate
Cytosol
Mitochondrion
Fatty acids
Acetyl CoA
ADP
Amino acids
CITRIC ACID CYCLE
ATP
CO_2
High-energy electrons and H^+
ADP
ELECTRON TRANSPORT SYSTEM
ATP
O_2 H_2O